Activity – Concept Check

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1. An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets.
   a. Find the first and the last address in the first subnet.
   b. Find the first and the last address in the last subnet.
   c. Find the number of addresses in each subnet.
   d. Find the subnet prefix.

2. Convert the following 32-bit IPv4 address from binary notation to dotted-decimal notation and vice versa:
   a. 11000000 10101000 00001010 01100101
   b. 10101100 00010000 00000101 11001010
   c. 168.110.121.12
   d. 126.92.56.110

3. Suppose a TCP connection is transferring a file of 9000 bytes. The first byte is numbered 15,000. The data are sent in five segments, where the first two segments carry 1500 bytes each, the next two segments carry 2000 bytes each, and the last segment carries 2000 bytes. Calculate the sequence number of the first byte in each of the five TCP segments.

4. Find the class ofthe following IP addresses.
   a. 208.34.54.12
   b. 238.34.2.1
   c. 114.34.2.8
   d.10101111 11000000 11110000 00011101
   e. 11011111 10110000 00011111 01011101
   f. 11101111 11110111 11000111 00011101

5. Find the netid and the hostid ofthe following IP addresses.
   a. 114.34.2.8
   b. 132.56.8.6
   c. 208.34.54.12

6. Change the following IPv4 multicast addresses to Ethernet multicast addresses. How many of them specify the same Ethernet address?
   a. 224.18.72.8
   b. 235.18.72.8
   c. 237.18.6.88
   d. 224.88.12.8

7. A router with IPv4 address 125.45.23.12 and Ethernet physical address 23:45:AB:4F:67:CD has received a packet for a host destination with IP address 125.11.78.10. Show the entries in the ARP request packet sent by the router. Assume no subnetting.

8. A host with IPv4 address 222.5.7.19 receives an IGMP query. When it checks its routing table, it finds two entries in its table: 227.4.3.7 and 229.45.6.23. What action should the host take? Should it send any messages? If so, what type and how many?

9. A live online gaming server normally sends 3 Mbps of data. During certain game events, the traffic can temporarily increase to 7 Mbps. To handle these sudden increases without allowing uncontrolled traffic into the network, a Token Bucket mechanism is implemented.
   The bucket can store a maximum of 12 tokens and tokens are generated at a rate of 3 tokens per second. Assume that one token is required to transmit one packet.
   (i) If packets are generated continuously, describe how the token bucket determines whether a packet can be transmitted immediately or must wait.
   (ii) During a sudden traffic burst, explain how the tokens accumulated in the bucket can be used to transmit packets at a rate higher than the normal rate.

10. A client has a packet of68,000 bytes. Show how this packet can be transferred by using only one UDP user datagram.